Quantile Balancing

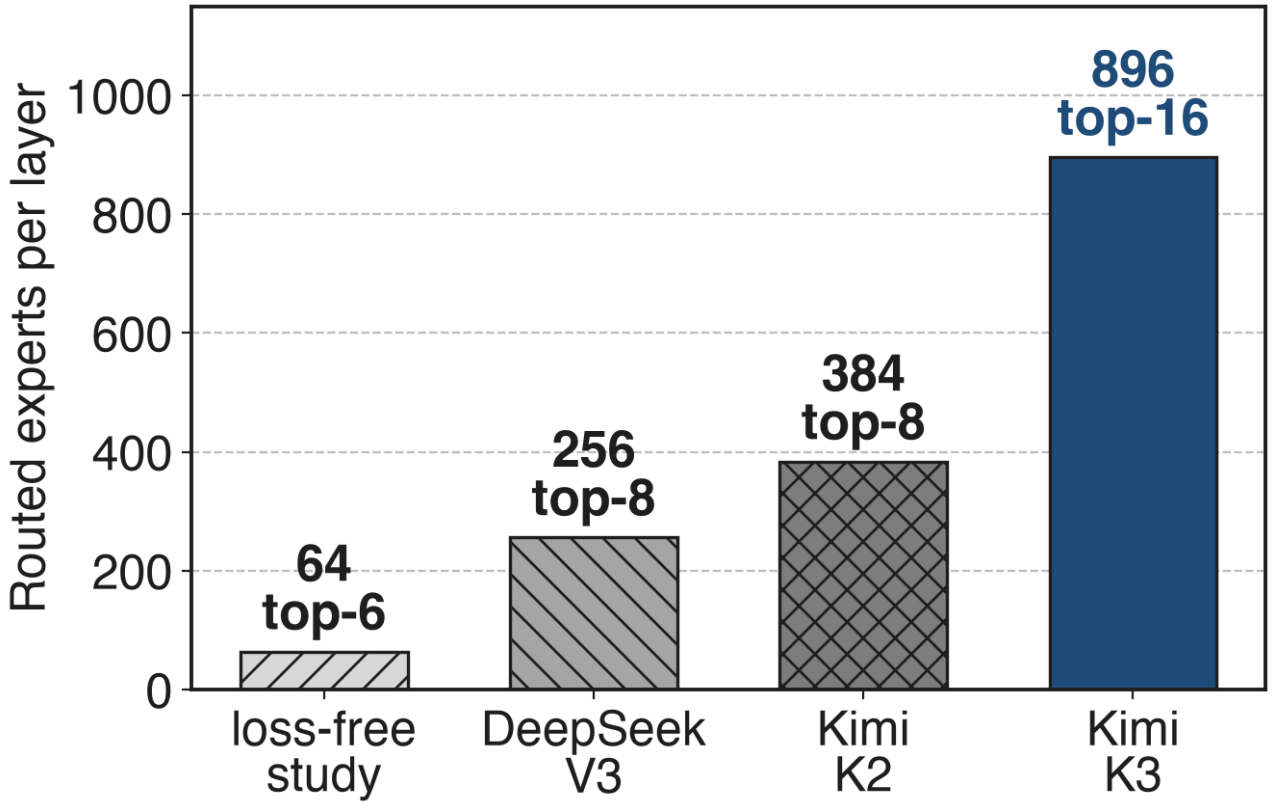
Set each MoE expert's routing bias from a score quantile

Moonshot AI · Kimi K3

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

K3 picks 16 of 896 experts per token

- Kimi K3: **2.8T** params, **104B** active
- Router score + per-expert **bias**; the bias only picks experts
- Every layer must keep **896** experts evenly loaded



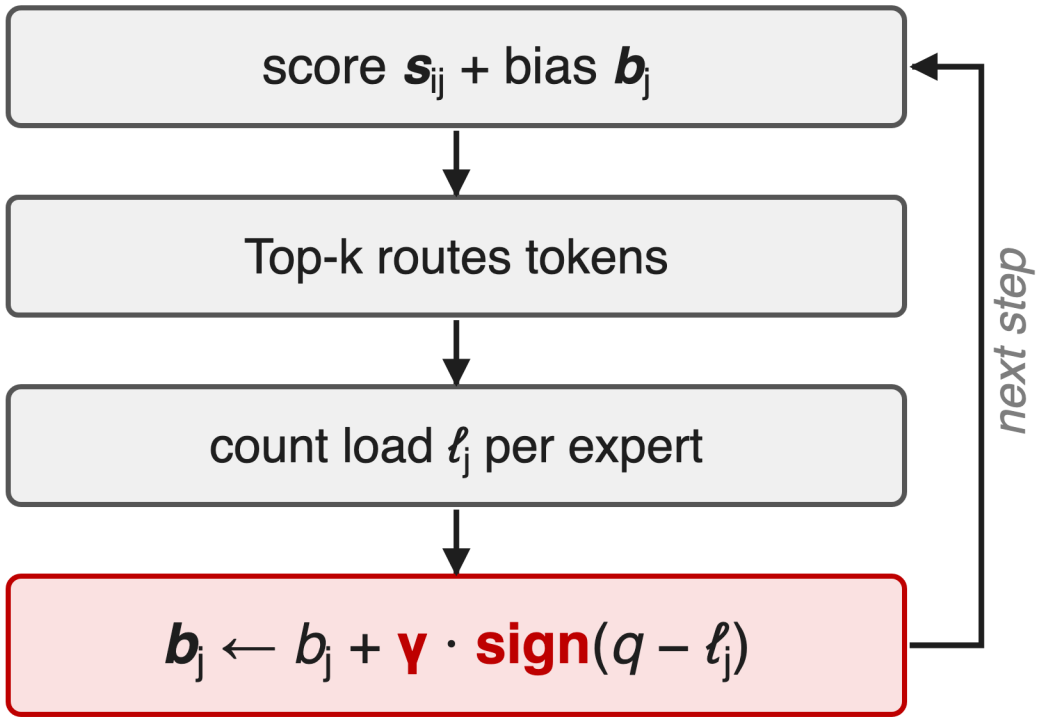
Sign updates leave hot experts overloaded

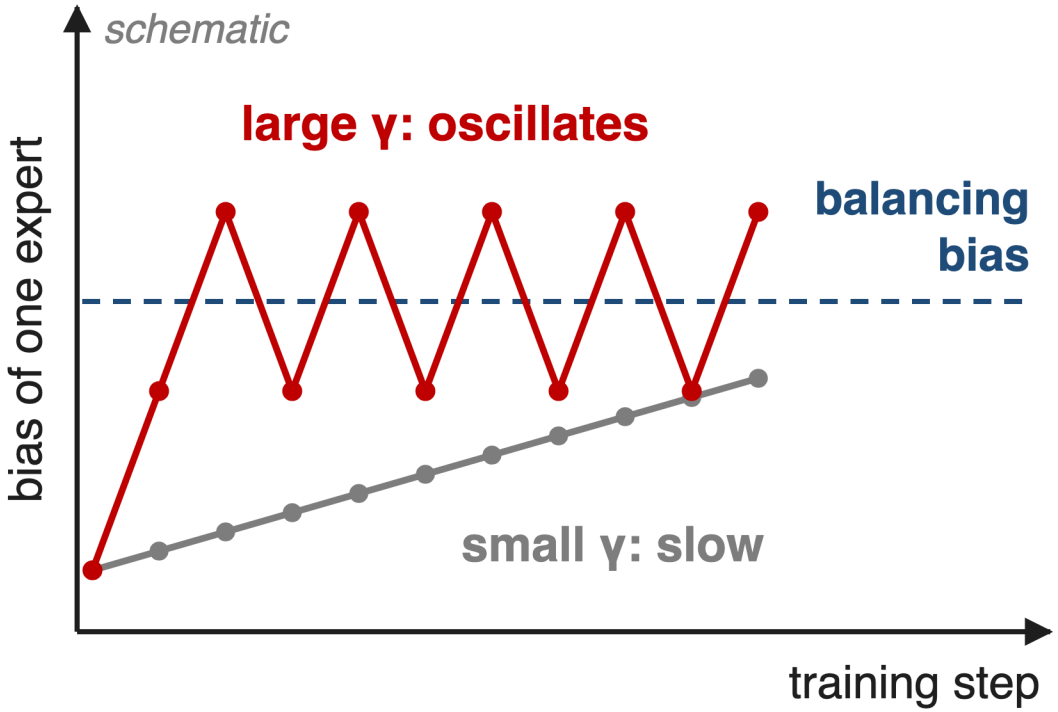
+34%

busiest expert's load above its fair share on an average batch: sign-update balancing, 64 experts (+13% at 16)

AvgMaxVio 0.34 vs 0.13; worst batch 2.71. BIP paper, arXiv 2502.15451, Tables 2–3 (0.3B / 1.1B MoE)

Loss-free balancing nudges each bias by ±γ

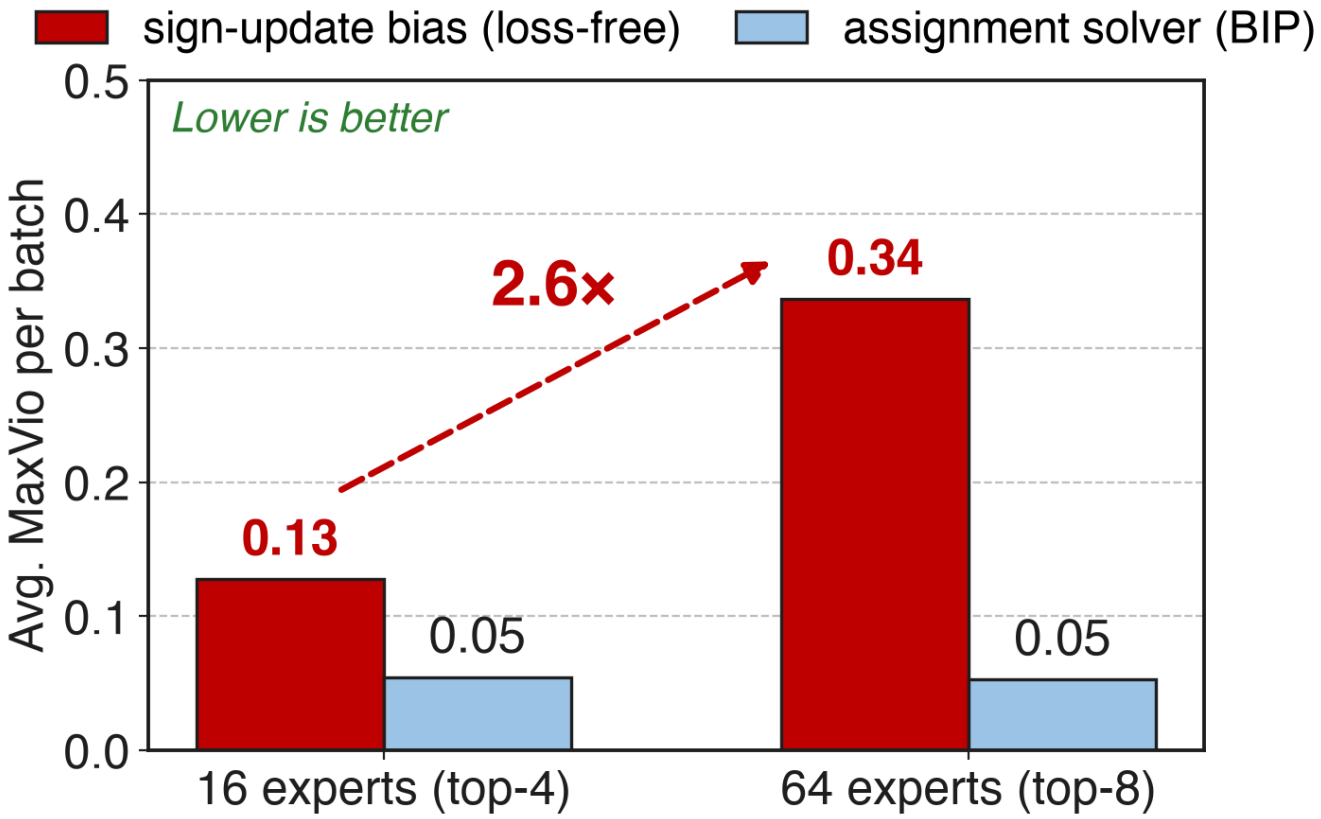




γ trades slow adaptation against oscillation, and one γ serves all 896 experts

Imbalance grows 2.6× from 16 to 64 experts

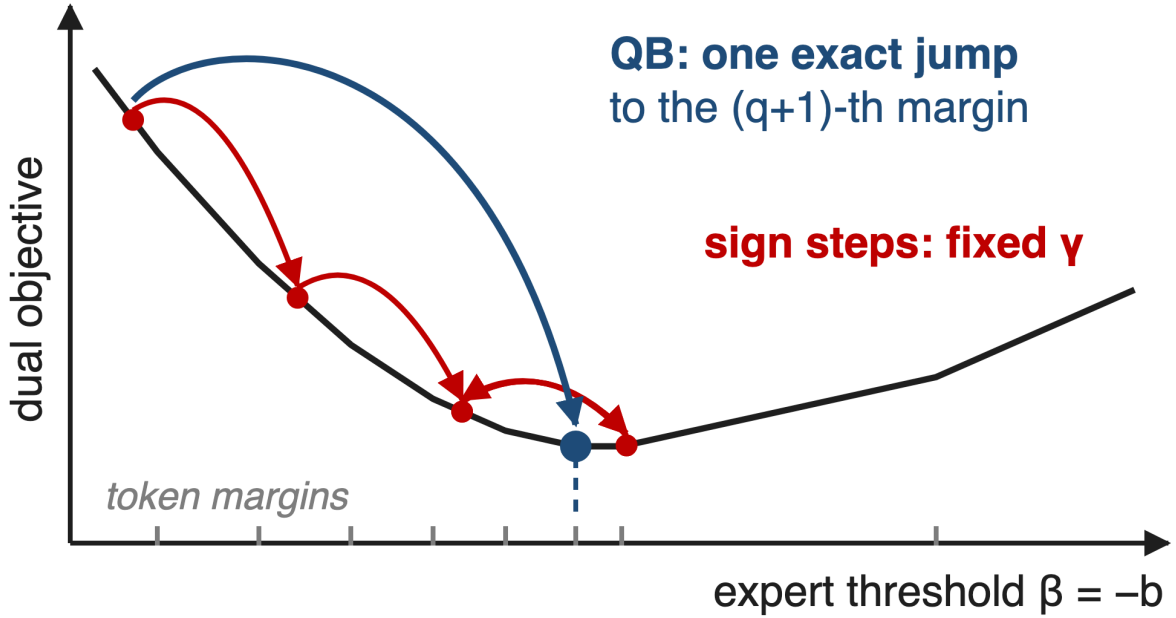
- Same sign update, **4×** the experts
- Assignment solver (BIP) stays at **0.05**
- K3 needs **14×** more: **896**



Third-party runs: Minimind MoE, 0.3B / 1.1B, softmax router (BIP paper, Tables 2–3)

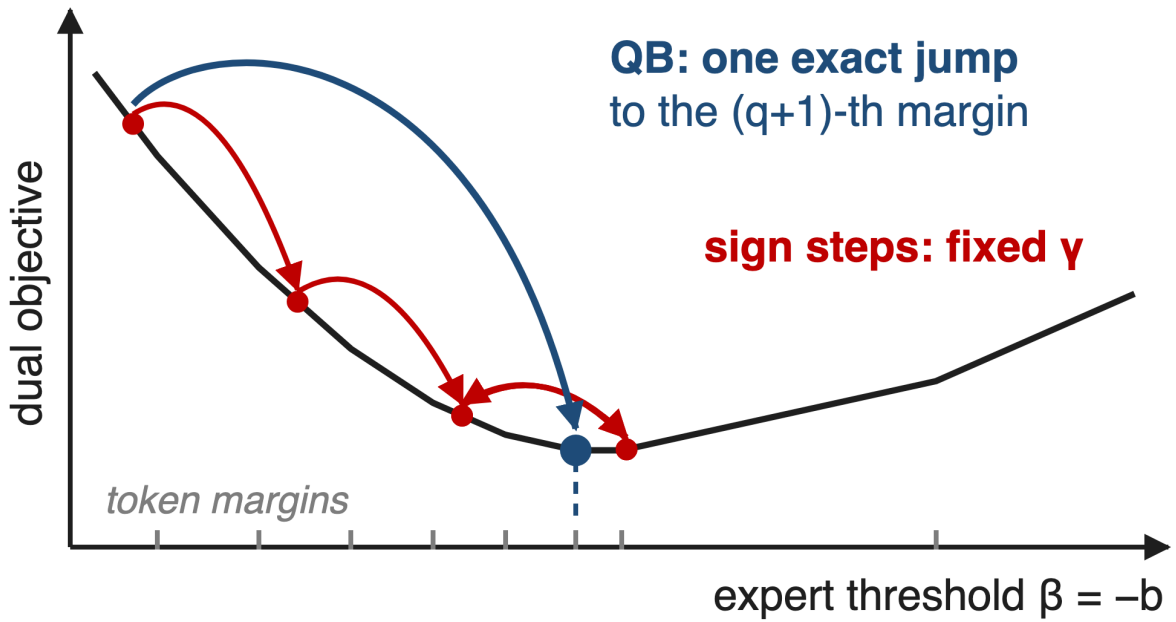
Insight: the balancing bias is a quantile

- Balanced Top-k = assignment LP
- Bias = expert's dual variable
- Slope = target – current load



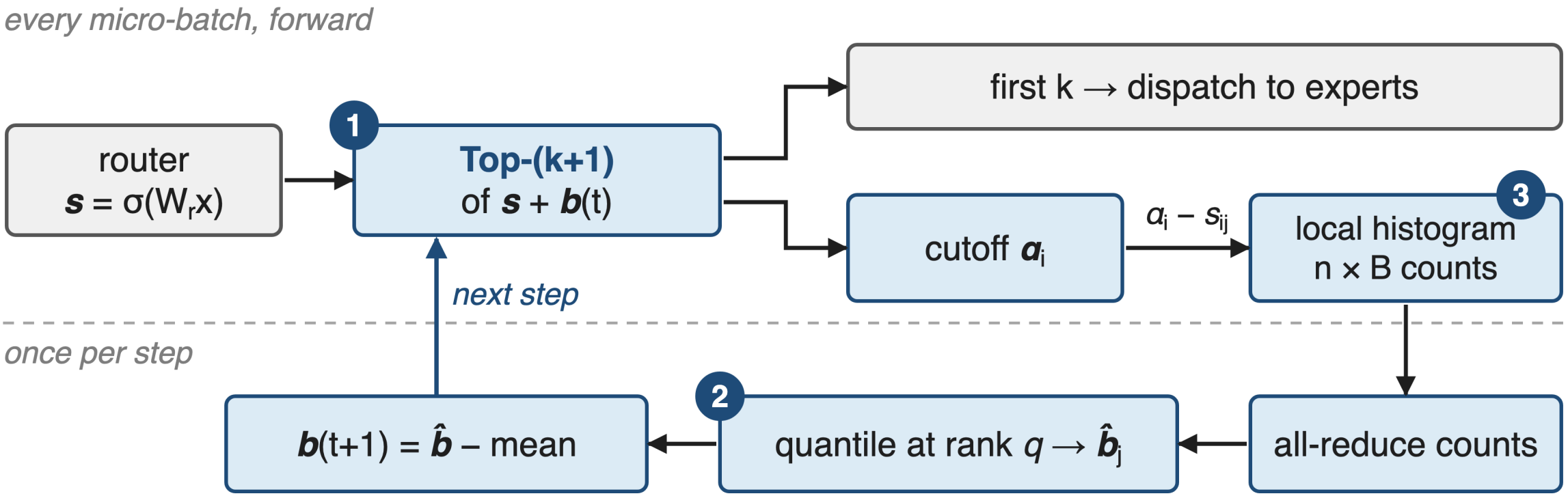
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Exact minimizer: the (1 – k/n)-quantile of the expert's margins

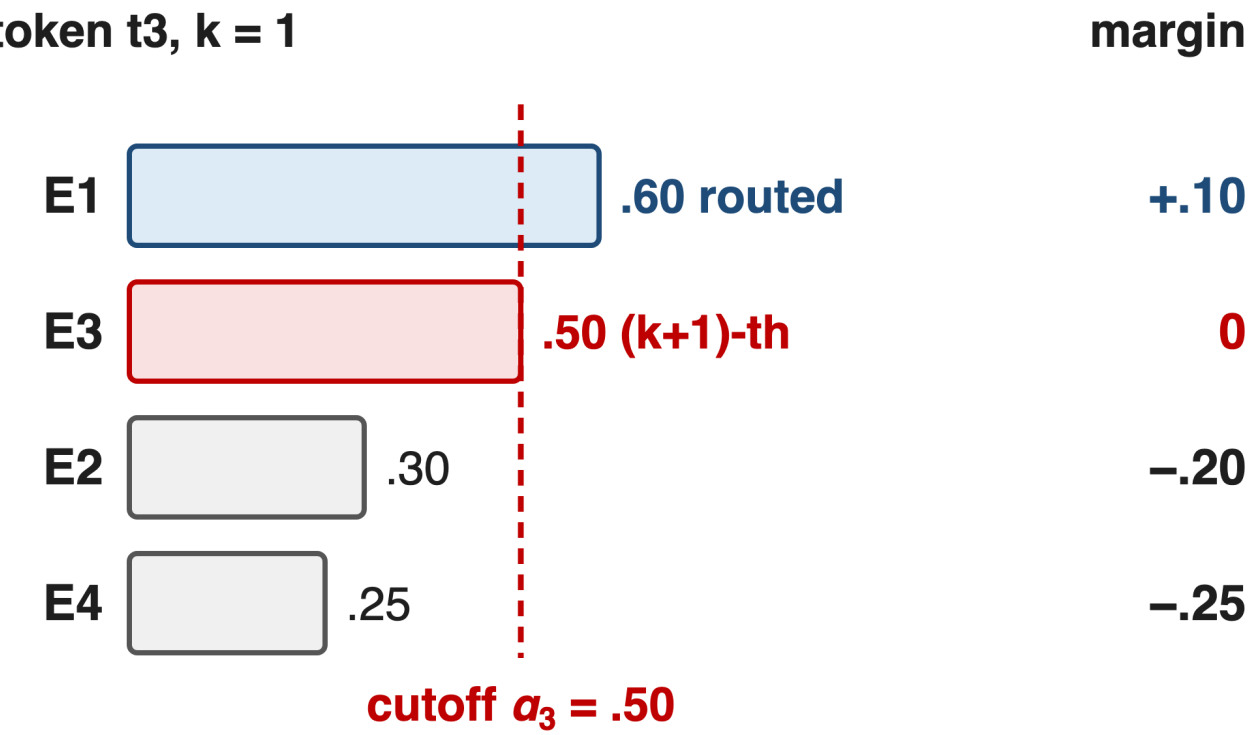
QB recomputes every bias once per step



❶ cutoff from Top-(k+1) · ❷ quantile → bias · ❸ histogram + one all-reduce. At inference the bias is frozen.

❶ Top-(k+1) gives each token a cutoff

- k routes plus **one** extra score
- Cutoff: the score an expert must beat
- Margin = score – cutoff



❷ Each bias jumps to its margin quantile

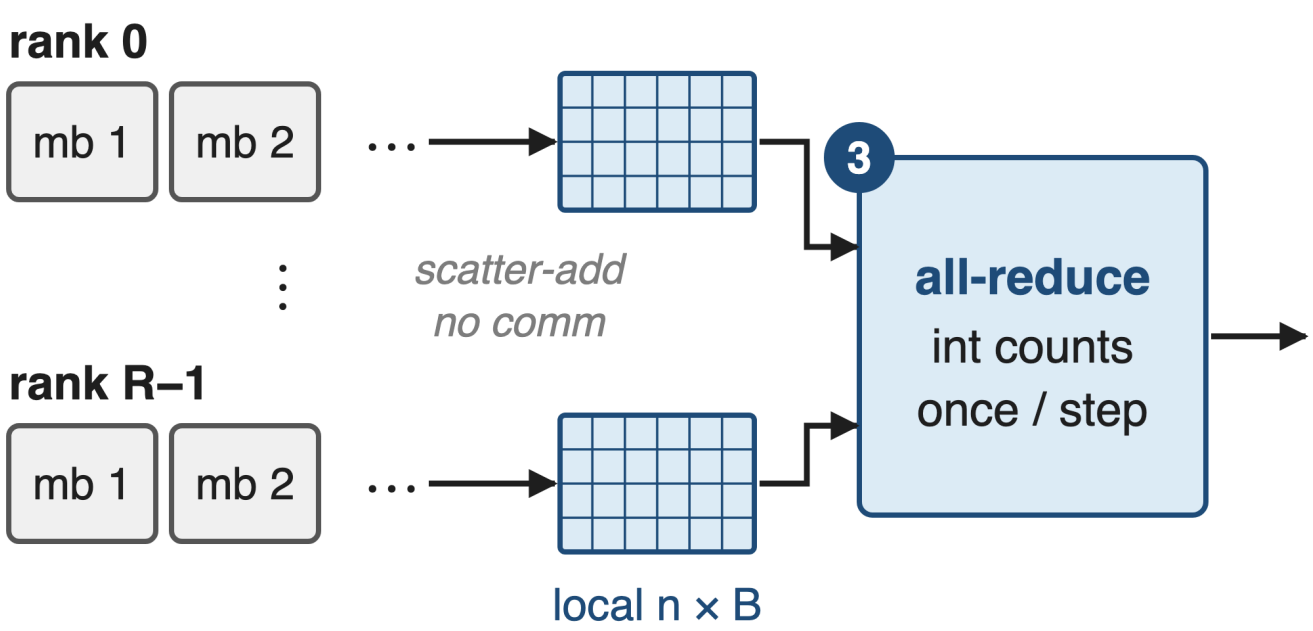
E1 · load 4		E2 · load 3		E3 · load 1		E4 · load 0	
t1	+.40	t5	+.35	t8	+.30	t4	0
t2	+.30	t6	+.25	t3	0	t7	0
t3	+.10	t7	+.08	t6	–.10	t8	–.15
t4	+.05	t1	0	t5	–.15	t3	–.25
⋯ 4 more		*⋯ 4 more*		*⋯ 4 more*		*⋯ 4 more*	

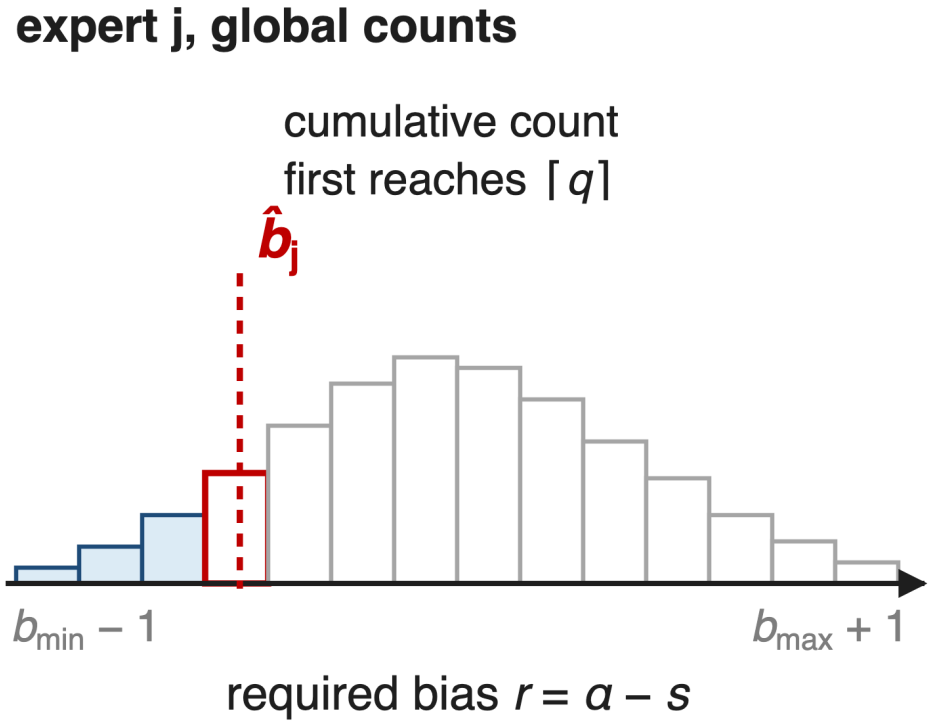
❷ Each bias jumps to its margin quantile

E1 · load 4 → 2		E2 · load 3 → 2		E3 · load 1 → 2		E4 · load 0 → 2	
t1	+.40	t5	+.35	t8	+.30	t4	0
t2	+.30	t6	+.25	t3	0	t7	0
t3	+.10	t7	+.08	t6	–.10	t8	–.15
t4	+.05	t1	0	t5	–.15	t3	–.25
⋯ 4 more		*⋯ 4 more*		*⋯ 4 more*		*⋯ 4 more*	
$\hat{b} = -\theta = -.10$		$\hat{b} = -\theta = -.08$		$\hat{b} = -\theta = +.10$		$\hat{b} = -\theta = +.15$	

Bias = –(q+1)-th largest margin, so exactly q = mk/n tokens clear it

❸ One all-reduce makes the quantile global

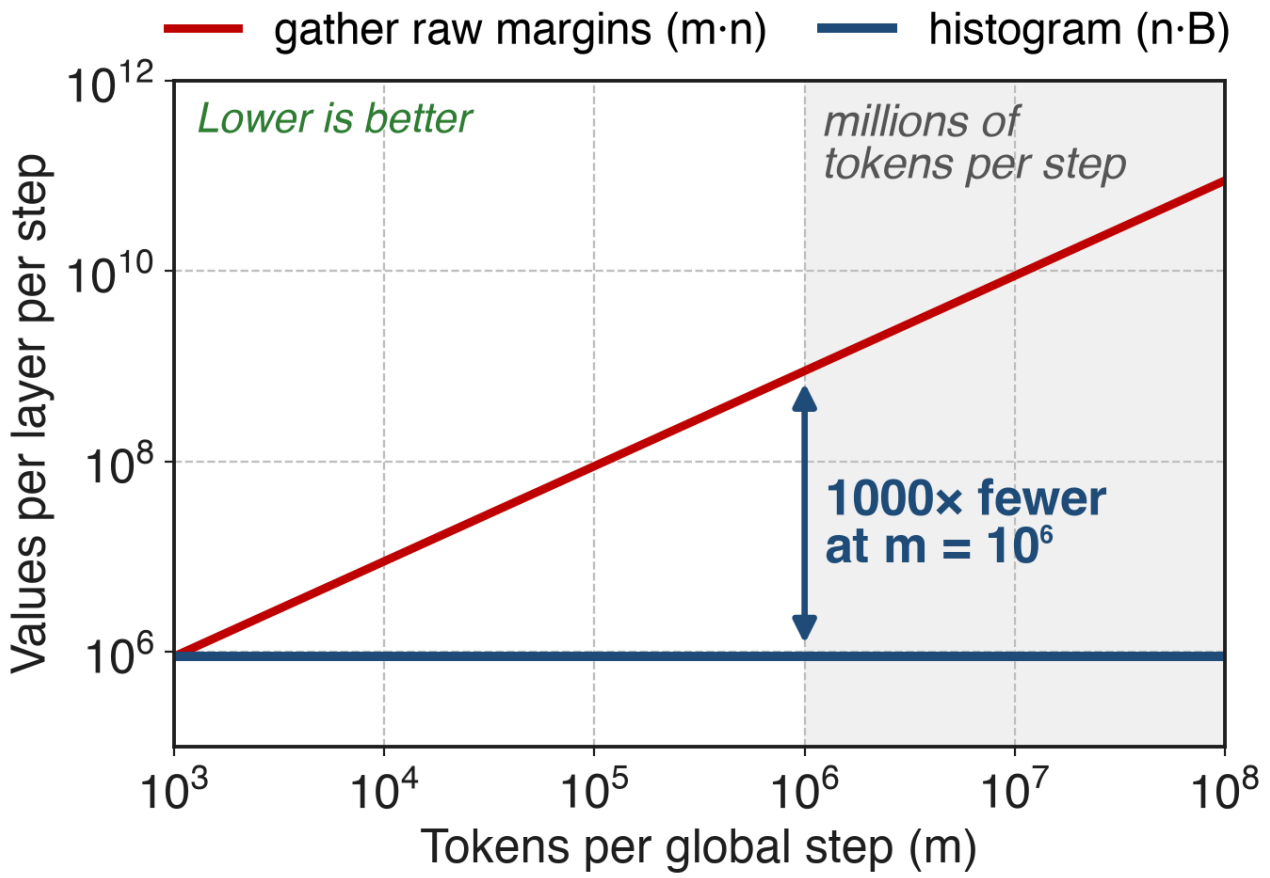




Error ≤ one bin: a few 10^{-3} at B = 1000; result independent of sharding

Quantile traffic drops below 1% of raw margins

- **n·B** counts per layer per step
- Same for any batch or sharding
- K3 measured **< 1%** of raw-margin exchange



Analytic: n = 896 experts, B = 1000 bins (upper bound; K3 text: a few hundred)

K3 balances 896 experts without a step size

- **QB**: router, per step
- No learning-rate-like knob
- No measurable residual imbalance

- **MoonEP**: EP ranks, per micro-batch
- Exactly S×K tokens per rank
- Deep dive: deck 27

QB keeps every expert evenly trained; MoonEP keeps every micro-batch's shapes static

The price: lag, two levels, no ablation

- **One-step lag**: step t's bias routes step t+1
- Needs Top-(k+1) and bounded scores
- Micro-batch skew still needs MoonEP
- **No public ablation**: no MaxVio or throughput vs sign update
- Inference: frozen bias, plain Top-k

Summary

- Sign updates: hottest expert **+34%** at 64 experts, **2.6×** worse than at 16
- **Insight**: the balancing bias is a quantile of routing margins
- Top-(k+1) cutoff, per-expert quantile, one histogram all-reduce
- K3: **896** experts, no step size, traffic **< 1%** of raw margins

https://arxiv.org/abs/2607.24653